HTML Tables

Presenting library and records information clearly and accessibly

Bachelor of Library and Information Studies
Bachelor of Records Management Systems

TITLE • AUTHOR

YEAR • STATUS

TYPE • ACCESS

What you will learn

The aim is a useful, understandable data table—not an elaborate website.

Structure

Use table, tr, th and td to organise rows, headers and data cells.

Explain

Add a caption, suitable header scope and grouped sections.

Apply

Create a short book catalogue or public records inventory using invented entries.

TAKEAWAY **Tables should be readable and meaningful before you add design decoration.**

Tables organise related data

A table supports comparison across records when fields repeat.

One entry per row

One book, public category or opening-hours entry is shown on one row.

One meaning per column

Values under the same header represent the same kind of information.

Two professional contexts

Book records and inventory entries use the same HTML table elements.

TAKEAWAY **Use tables when people need to compare related values.**

When a table is appropriate

Choose a table because the information is tabular.

Good example: library catalogue

Title, author, publication year and availability are related columns.

Good example: public inventory

Category, fictional reference, year and access status can be compared.

Not a layout technique

Menus, pictures and whole-page sections should not be arranged with data tables.

TAKEAWAY **A table represents meaning; CSS will handle general webpage layout.**

Rows and columns

Read across a row to understand one item; read down a column to compare items.

Book	Author	Status
Digital Collections	N. Juma	On loan
Information Services	A. Mrema	Available
Archives	P. Lema	Available

Row

One book entry across the table.

Column

One repeated field, such as author.

Cell

The intersection holding one value.

TAKEAWAY **Count the heading row as a row.**

The simplest HTML table

A table is a container containing rows; rows contain individual cells.

table

Outer wrapper for the related information.

tr

Creates each horizontal row.

td

Creates an ordinary data cell.

HTML EXAMPLE

```
<table>
  <tr>
    <td>Book</td>
    <td>Status</td>
  </tr>
  <tr>
    <td>Digital Libraries</td>
    <td>Available</td>
  </tr>
</table>
```

TAKEAWAY **The example is a two-row, two-column table.**

From table to row to cell

Elements must be nested in the correct order.

table → tr → td

A data cell goes inside a row, not directly inside the table.

Opening and closing tags

Close each td, tr and table element in the right order.

Matching fields

Simple data rows normally contain the same number of cells.

TAKEAWAY **Correct nesting is more important than styling.**

Understanding the tr element

Every separate record needs its own tr.

Heading row

First row identifies the columns.

Data rows

Following rows hold the actual entries.

Another item

Add another complete tr, not extra td outside a row.

HTML EXAMPLE

```
<tr>
  <td>Information Services</td>
  <td>Available</td>
</tr>
<tr>
  <td>Digital Libraries</td>
  <td>On loan</td>
</tr>
```

TAKEAWAY **Each tr forms one horizontal record.**

Data cells with td

Use td for an actual value rather than its heading.

Value

Each cell carries a fact about one item.

Order

Cell order follows the header order.

Use correct content

A status cell should contain a status, not an author.

HTML EXAMPLE

```
<tr>
  <td>Digital Libraries</td>
  <td>N. Juma</td>
  <td>2024</td>
  <td>Available</td>
</tr>
```

TAKEAWAY **One td corresponds to one field value.**

Column headers with th

Header cells explain how to interpret the values below.

Meaning

Author is a label, not an author name.

Semantics

th identifies a header to browsers and assistive technology.

Orientation

Add scope="col" to explicitly label column data.

HTML EXAMPLE

```
<tr>
  <th scope="col">Title</th>
  <th scope="col">Author</th>
  <th scope="col">Year</th>
  <th scope="col">Status</th>
</tr>
```

TAKEAWAY **Use th for labels rather than making td text merely bold.**

A catalogue with useful headings

Visible headers make comparisons faster and more accurate.

Title	Author	Year	Status
Digital Libraries	N. Juma	2024	Available
Recordkeeping	T. Komba	2023	On loan
Cataloguing	A. Mrema	2022	Available

Heading

What does each value describe?

Row

Which details belong to one title?

Status

Can a reader identify a book to borrow?

TAKEAWAY **Each column has a consistent meaning across all rows.**

Create a meaningful caption

Place a short caption directly after the opening table element.

Name the table

Describe the dataset: e.g. sample community books.

Understand before reading

The caption explains what the reader can expect.

Accessibility

Captions help readers locate the appropriate table.

HTML EXAMPLE

```
<table>
  <caption>Community Library: Sample
Books</caption>
  <tr>
    <th scope="col">Title</th>
    <th scope="col">Status</th>
  </tr>
  ...
</table>
```

TAKEAWAY **A good caption names the data without repeating every column heading.**

Use scope="col"

scope=col associates a header with values in its column.

Header

A cell such as Publication year is a column heading.

Relationship

Values below it should contain years.

Practical benefit

The relationship is clear to readers and assistive technology.

HTML EXAMPLE

```
<thead>
  <tr>
    <th scope="col">Title</th>
    <th scope="col">Year</th>
  </tr>
</thead>
```

TAKEAWAY **Header scope describes relationships, not colours.**

Use scope="row"

Some tables use a row label such as Monday or Annual reports.

Row label

The first cell names the row.

Related value

The adjacent cell contains its corresponding information.

Example

Monday is paired with opening hours, not a different day.

HTML EXAMPLE

```
<tr>
  <th scope="row">Monday</th>
  <td>08:00–17:00</td>
</tr>
<tr>
  <th scope="row">Saturday</th>
  <td>09:00–13:00</td>
</tr>
```

TAKEAWAY **scope=row helps explain data beside the header.**

A table with both header directions

Column and row headers can work together in the same table.

Day	Open	Close
Monday	08:00	17:00
Friday	08:00	16:00
Saturday	09:00	13:00

Columns

Open and Close describe time values.

Rows

Monday, Friday and Saturday identify days.

Relationship

A time belongs to a particular day and column.

TAKEAWAY **Rows and columns should identify the meaning of every data cell.**

thead, tbody and tfoot

Use group elements to distinguish headings, main data and summaries.

thead

Contains column heading rows.

tbody

Contains most of the records or data entries.

tfoot

Contains a total or summary only when useful.

HTML EXAMPLE

```
<table>
 <thead><tr><th>Type</th></tr></thead>
 <tbody><tr><td>Books</td></tr></tbody>
 <tfoot><tr><td>Total: 24</td></tr></tfoot>
</table>
```

TAKEAWAY **A table does not need tfoot unless it has a meaningful summary.**

Why grouping helps

Table groups improve readability of the HTML source and the intended structure.

Header group

A reader or author can locate the column labels in thead.

Data group

The repeated item rows belong together inside tbody.

Optional totals

Include a summary only when it tells the reader something useful.

TAKEAWAY **Table groups organise rows; they do not create new columns.**

Library catalogue example

One book per row, with four consistent fields.

Title	Author	Year	Status
Library Services	A. Mrema	2022	Available
Digital Libraries	N. Juma	2024	On loan
Cataloguing	P. Lema	2021	Available

Purpose

Show availability of example titles.

Code pattern

Caption, header row, and data rows.

Privacy

Fictional books and authors only.

TAKEAWAY **An HTML catalogue illustrates data; it does not yet provide a live lending database.**

Understanding catalogue fields

A useful catalogue table shows related information in one place.

Title

Identifies the publication or information resource.

Author and year

Describe the publication and help disambiguate entries.

Status

Shows fictional availability; real-time status would require a connected system.

TAKEAWAY **A demonstration catalogue is not automatically a live OPAC.**

Records inventory example

A public inventory can describe non-sensitive categories and access information.

Category	Reference	Year	Access
Annual reports	DEMO-001	2023	Public
Service guides	DEMO-002	2024	Public
Policies	DEMO-003	2025	Public

Reference

Use only invented numbers for training.

Access

Use generic public status; no restricted files.

Usefulness

Visitors can see which public items exist.

TAKEAWAY **Never publish actual protected records to demonstrate tables.**

Privacy in records-centre tables

A properly formatted table is not automatically safe to publish.

Safe practice

Fictional examples and general information about public services.

Unsafe data

Actual patient records, personnel files or confidential references.

Professional responsibility

Table content must follow the institution’s information-access rules.

TAKEAWAY **Content protection is a separate requirement from HTML presentation.**

Cells spanning two columns

colspan gives one cell the width of two columns.

Use case

A category label needs to stretch across two fields.

Count cells

A colspan=2 cell takes the place of two ordinary cells.

Keep it simple

Avoid spanning just for appearance.

HTML EXAMPLE

```
<table>
  <tr>
    <th colspan="2">Library contact</th>
  </tr>
  <tr>
    <th scope="row">Office</th>
    <td>Reading Hall</td>
  </tr>
</table>
```

TAKEAWAY **A row with colspan="2" may legitimately contain just one cell.**

Cells spanning two rows

rowspan allows one label to apply across vertically adjacent rows.

Meaning

One area has multiple related services.

Syntax

rowspan="2" covers the current and following row.

Avoid duplicates

Do not repeat the spanning cell in the next row.

HTML EXAMPLE

```
<table>
  <tr>
    <th rowspan="2">Library</th>
    <td>Reference help</td>
  </tr>
  <tr>
    <td>Reading space</td>
  </tr>
</table>
```

TAKEAWAY **Spanning must match the intended data relationship.**

Spanning cells carefully

Complex cell arrangements create extra reading work.

Plan before coding

Sketch the grid and count which cells span other positions.

Use consistent headers

Keep headings associated with their intended values.

Prefer simple tables

When a table becomes confusing, consider separate smaller tables.

TAKEAWAY **Avoid unnecessary merged cells in beginner tables.**

Avoid layout tables

Tables are for data rather than positioning unrelated page sections.

Not for site layout

A sidebar, menu and banner do not form tabular data.

Better approach

Use semantic HTML regions such as header, nav and main.

Styling later

CSS will create columns, spacing and responsive layouts.

TAKEAWAY **Use the correct HTML element for the information’s meaning.**

Missing and mismatched tags

Small errors can put values under the wrong headings.

Missing closing cell

Close every td and th correctly.

Wrong number of cells

Count columns and check each repeated record row.

Forgotten tr

Start a new tr for each new book or inventory entry.

TAKEAWAY **When a table looks wrong, check tag nesting before colours.**

How browsers display plain tables

A table can work correctly before you add CSS.

HTML establishes meaning

Rows, headings and values already form a valid data grid.

Browser appearance varies

Borders, spacing and text sizes depend on default styles.

CSS improves presentation

A stylesheet can provide borders, padding and readable backgrounds.

TAKEAWAY **HTML first; visual styling second.**

Optional CSS for readability

A small stylesheet can make an existing HTML table easier to scan.

border-collapse

Combines adjacent borders.

padding

Adds space inside each cell.

background

Gives column headers a visual distinction.

HTML EXAMPLE

```
<style>
  table {
    width: 100%;
    border-collapse: collapse;
  }
  th, td {
    border: 1px solid #bbc9c8;
    padding: 10px;
  }
  th { background: #e8f3ef; }
</style>
```

TAKEAWAY **Styling does not replace proper th or caption elements.**

Narrow screens and many columns

A table with too many columns can be hard to read on a phone.

Keep essential fields

Use clear, short column labels and meaningful values.

Allow scrolling

Keep the table structure while enabling horizontal movement.

Simplify complex information

Split unrelated datasets into separate smaller tables.

TAKEAWAY **Accessibility is part of good design, not an optional decoration.**

Build and test a table

Complete the practical by checking both HTML and what the browser shows.

Start with a sample

Choose a library catalogue or public records inventory.

Edit the structure

Provide a caption, four column headers and fictional data rows.

Check the result

Open the file, add rows and make sure values align with headings.

TAKEAWAY **Save the HTML source and verify it in a browser.**

Mini practice: add a new row

A simple new entry strengthens your understanding of cells and rows.

Library track

Add a fictional fourth book with title, author, year and status.

Records track

Add a fictional fourth public category with reference, year and access.

Review

Count four values; check the caption and headers still make sense.

TAKEAWAY **A new entry is a complete tr containing matching cells.**

Unit 5: essential takeaways

You can now present simple structured information in HTML.

Main structure

table contains tr; rows contain th and td.

Meaning and access

caption, scope and grouping make data easier to understand.

Next topic

Unit 6 introduces HTML forms: labels, controls and basic validation.

TAKEAWAY **Use tables for meaningful data and keep all practice records fictional.**